In **competitive programming**, success comes down to milliseconds. Every fraction of a second in execution time determines the difference between victory and defeat. For nine years, I obsessed over crafting optimal solutions, pushing my code to run at lightning speed. Working with data structures like segment trees and Fenwick trees showed me that performance is not just about better algorithms, it is also about how we organize and access our data. This insight drove my interest in data management systems.

This realization became even clearer during my first research collaboration with teams from the **University of Lorraine** and **University of Malta** on the *SemEval-2023 Task 2: Fine-grained Multilingual Named Entity Recognition (MultiCoNER 2)* competition. We developed **garNER**, a system designed to retrieve and integrate knowledge from Wikipedia's database to improve the training dataset. Unlike methods that focus on refining machine learning models, our innovation was in how we **managed and processed data**. We dynamically queried the Wikipedia API to gather relevant information and enriched the training data with detailed context. This approach combined data augmentation with predictions from transformer models like XLM-RoBERTa and mBERT. It demonstrated that effective data management and augmentation are crucial for improving multilingual entity recognition.

The main challenge was handling large-scale multilingual data efficiently while ensuring the retrieved information truly improved the system's performance. To address this, we carefully designed a pipeline to manage data retrieval, preprocessing, and integration. This focus on efficient data handling proved crucial to our success, enabling us to be **the only team** to rank in the top 10 across 10 language tracks. More importantly, it showed that strategic **data management** is just as important as model architecture in improving AI systems. This experience solidified my interest in developing advanced techniques for data processing and integration to solve real-world problems.

Building on my experiences with efficient data management, I explored real-time interaction systems during my undergraduate thesis. I worked under the supervision of **Dr. Khalad Hasan** from the **University of British Columbia**, **Dr. Farida Chowdhury** (now at BRAC University) and **Dr. Md Forhad Rabbi** from my university. This work was published in **EAI MobiQuitous 2023** [2]. I designed a smartphone interaction system where users could control an on-screen cursor through **head movement tracking** to select items from hierarchical marking menus. A key technical problem in this work was efficiently processing and responding to continuous streams of face landmark data while maintaining responsive interaction. The system had to process multiple facial landmarks per frame in real time, which could cause lag if the data was not managed efficiently.

In my professional experience, I have worked on implementing advanced **information retrieval systems**, focusing on heterogeneous document processing and multi-modal content integration. One of my key projects involved building a scalable asynchronous pipeline for processing diverse data sources, including structured documents, web content, and multimedia transcripts, while ensuring cross-format information coherence. Using **Pinecone** as the vector database, I designed a distributed chunking architecture integrated with embedding models and

vector search, emphasizing hierarchical document representation and precise source attribution.

This system incorporated dynamic metadata enrichment and context-aware retrieval mechanisms, addressing real-world challenges in cross-domain knowledge integration. Through this work, I explored practical solutions to optimize semantic chunking, improve provenance-aware response synthesis, and enhance retrieval efficiency at scale. The experience of working on production **Retrieval Augmented Generation (RAG)** systems deepened my understanding of the complexities involved in large-scale information retrieval. It also inspired several research interests, such as dynamic chunk size optimization based on semantic coherence and strategies for aligning cross-modal information.

My fascination with data retrieval systems led me to develop FindResearch.online, a platform for discovering academic papers across diverse sources like CORE, Crossref, Papers with Code and arXiv. The system combines **BM25-based** relevance sorting with **DistilBERT** for contextual insights, chosen for its lightweight design that enabled deployment alongside the JavaScript backend, offering AI-powered services at **no cost** to users. To handle diverse academic formats and ensure index freshness, I implemented batch processing strategies and optimized index structures, enabling consistent query performance as the dataset grew. Serving over **1,000** monthly users, this project deepened my understanding of balancing advanced ranking algorithms, data management, and accessible deployment in modern information retrieval systems.

Throughout this journey, I have maintained my commitment to competitive programming, qualifying for high-level competitions including the **International Collegiate Programming Contest Asia West Championship**. What began as a high school interest evolved into nearly a decade of dedicated training in algorithms and data structures. Through these years, I reached the top **0.03%** on Codeforces and top **0.01%** on CodeChef, platforms with **over a million** users each. This extensive practice has significantly influenced how I approach system design and optimization, developing my ability to analyze complex systems methodically and investigate performance bottlenecks systematically.

Beyond my own technical pursuits, I am passionate about empowering others to excel in computer science. I have served as a competitive programming trainer at International Islamic University Chittagong and an instructor for the online education platform RoboAdda. This passion for teaching extended beyond formal roles. I coordinated our university's programming training camps and was invited as an industry expert to deliver lectures on Software Engineering at **multiple universities** under a **government** initiative funded by the **World Bank**. Beyond traditional teaching, I maintain a **technical blog**. Observing the scarcity of Bengali technical content, I launched a **YouTube channel** too. I also run a **Discord** community where students can get personalized help with competitive programming problems. Through these diverse teaching platforms, I've reached thousands of Bengali-speaking students who previously had limited access to technical education. Breaking down complex algorithmic concepts for others has deepened my appreciation for both theoretical foundations and practical data

management challenges. This is precisely the intersection where I aim to contribute through research.

I am eager to pursue my Ph.D. at the University of Utah under the guidance of **Dr. Anna Fariha**. Her focus on enhancing the usability and trustworthiness of data systems aligns closely with my passion for creating efficient yet user-friendly tools, as demonstrated in my work on FindResearch.online and Retrieval-Augmented Generation pipelines. Dr. Fariha's emphasis on applying theoretical algorithms to practical challenges resonates with my **competitive programming** background, where I developed the ability to analyze complex problems systematically and design efficient solutions. The University of Utah offers the ideal setting to advance data management research, combining strong theoretical foundations with meaningful real-world impact. By contributing to the advancement of user-friendly and efficient data systems, I aspire to address fundamental challenges in data management and create solutions with broad societal impact.

References

[1] garNER at SemEval-2023: Simplified Knowledge Augmentation for Multilingual Complex Named Entity Recognition. SemEval 2023 (SemEval-2023) (pp. 823-835). Link
[2] Exploring Optimal Placement of Head-based Hierarchical Marking Menus on Smartphones. EAI MobiQuitous 2023 (pp. 381-392). Link